ROLL NO: _____

# MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

## END-SEM EXAMINATION — DECEMBER / JUNE 2024

FIRST YEAR (COMMON TO ALL BRANCHES) — SEMESTER 2

**Time Allowed: 3 Hours** **Maximum Marks: 70**

*NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.*
*2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].*
*3. Assume suitable missing data, if any, and state it clearly.*
*4. Use of scientific non-programmable calculator is permitted.*

---

**--- UNIT I ---**

**Q.1** Solve Second Order Linear Differential Equations with constant coefficients using Method of Variation of Parameters. [14 Marks]

**[ OR ]**

**Q.1 (Alt)** Solve Cauchy-Euler Homogeneous Linear Differential Equations. [14 Marks]

**--- UNIT II ---**

**Q.2** Find Laplace Transform of standard functions and Periodic functions. State First and Second Shifting Theorems. [14 Marks]

**[ OR ]**

**Q.2 (Alt)** Evaluate Inverse Laplace Transform using Convolution Theorem. Solve Initial Value Problems. [14 Marks]

**--- UNIT III ---**

**Q.3** Form Partial Differential Equations (PDE) by eliminating arbitrary constants and functions. [14 Marks]

**[ OR ]**

**Q.3 (Alt)** Solve first-order linear PDE using Lagrange's Method: Pp + Qq = R. [14 Marks]

**--- UNIT IV ---**

**Q.4** Solve 1D Heat Conduction and 1D Wave Equations using Method of Separation of Variables. [14 Marks]

**[ OR ]**

**Q.4 (Alt)** Solve Laplace Equation in two dimensions under given boundary conditions. [14 Marks]

**--- UNIT V ---**

**Q.5** Explain Fourier Transform, Sine and Cosine transforms and their inversion formulas. [14 Marks]

**[ OR ]**

**Q.5 (Alt)** Solve Boundary Value Problems using Fourier Transform techniques. [14 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****